Case Investigation Brief: Fertiliser Token

Team [NAME] | AI Synergy Hackathon 2026 | KhadSetu: a token should be a promise, not a queue number.

- Track: [your track, e.g. AI in Agriculture / AI for All]
- Case: a farmer must get a digital token before buying fertiliser; today the token is a queue position, not a guarantee.
- This brief states what we know, what we assume, and what we still must verify.

1. Selected theme / case

Fertiliser Token: a farmer must obtain a digital token before buying fertiliser. The token is a queue position and not a guarantee of stock, date or distance. We build KhadSetu, which issues a token only after stock is reserved at a specific centre, on a date before the sowing deadline, at the nearest feasible centre.

2. Stakeholders

Stakeholder	Wants	Pain today	Influence
Landowner farmer	Fertiliser before sowing	Travel, waiting, stock-outs, wrong dates	Low / high collectively
Tenant / sharecropper	Same access	No Farmer ID: no token at all	Very low
Centre operator	Orderly counters, correct stock	Crowds, disputes, no forecast	Medium
Village verifier	Fair, defensible decisions	Pressure, no clear process	Medium
District officer	Working system, no unrest	No live demand-vs-stock view	High
State fertiliser dept	Policy that holds	Rollout suspended in ~3 months	Highest
Suppliers / logistics	Predictable demand	Unclear where stock should go	Medium (upstream)

3. Problem identified

Symptoms: 30-50 km travel, hours of waiting, tokens dated after the sowing need, stock gone on arrival, repeat trips, some farmers shut out.

Root causes: (1) tokens issue without checking stock; (2) dates come from queue position, not sowing deadlines; (3) no matching by distance and stock; (4) Farmer-ID gate excludes tenants and one OTP failure blocks access; (5) a 3-district pilot went statewide without scale testing. Separately: a total supply shortage is NOT solved by tokens.

4. Alternative problem statements considered

Scored 1-5 (5 best). D totals highest on feasibility, yet we choose A: D never fixes the root cause (unbacked tokens), and A produces the demand signal B needs later.

Opt	Description	Imp	Feas	AI	Data	Risk	Tot
A*	Stock-backed smart token (chosen)	4	4	3	3	3	17
B	Upstream stock logistics	5	2	4	2	2	15
C	Tenant identity layer only	3	4	2	3	3	15

Opt	Description	Imp	Feas	AI	Data	Risk	Tot
D	Info app (stock display + SMS)	3	5	3	3	4	18
E	Village/doorstep delivery redesign	5	1	1	2	1	10

5. Evidence and assumptions

Status key: BRIEF = from the case brief; ASM = our stated assumption; VERIFY = needs a real source before it is presented as fact.

Claim	Source	Confidence	Status
30-50 km travel, hours of waiting	Case brief	Medium (anecdotal)	BRIEF
Token ignores stock/date/distance	Case brief	High	BRIEF
Tenants lack Farmer ID	Case brief	High	BRIEF
One OTP failure can block	Case brief	Medium	BRIEF
3-district pilot, statewide, suspended ~3 months	Case brief	High (fact); cause unknown	BRIEF
Small/marginal holding share (India)	Agri Census	-	VERIFY
Tenant cultivator share	NSS / Census	-	VERIFY
Fertiliser subsidy size	Union Budget	-	VERIFY
Rush-week demand clustering	Assumption	Medium	ASM
OTP fail 15%, no-show 10%	Assumption	Low	ASM: replaced by pilot data

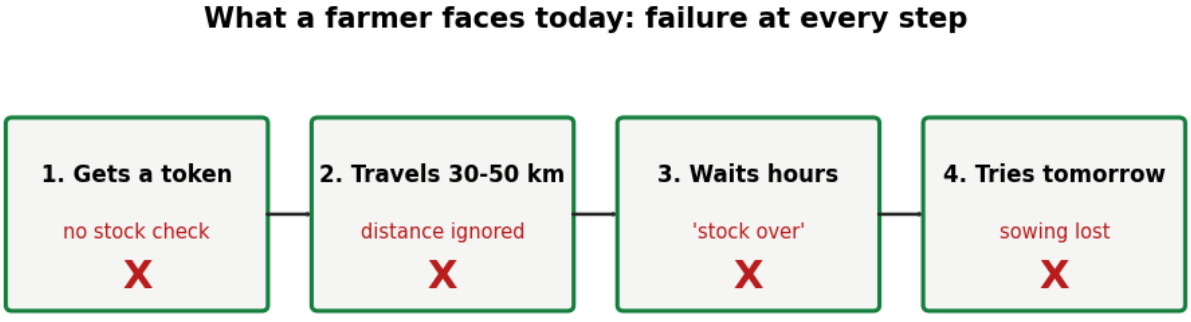
Unknown: real per-centre stock; true suspension cause; exclusion counts. We state the suspension as fact, never its cause.

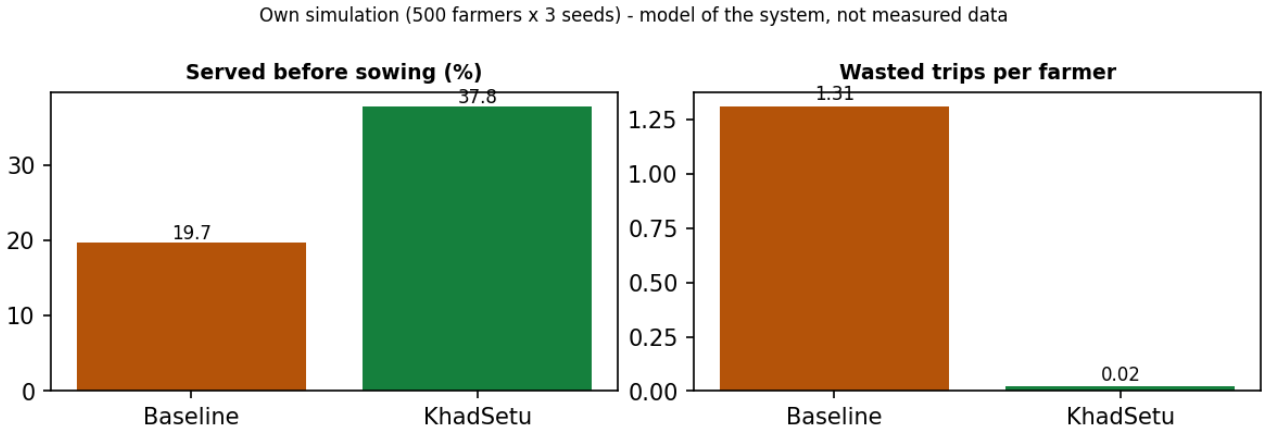
6. Why we chose this problem

It fixes the root cause behind every symptom; it is buildable and testable in 36 hours; the simulator gives evidence instead of claims; AI has a real but bounded role; and it creates the demand data option B needs later. D was simpler and scored higher on feasibility, but information without reservation leaves the core failure intact.

7. Proposed solution (with measured results)

KhadSetu reserves stock first, matches the sowing date and nearest feasible centre, waitlists honestly with ETAs, includes tenants/OTP failures via audited fallbacks, and tests rules in a simulator. AI helps state needs, forecasts, flags anomalies and explains; rules decide stock, humans decide eligibility. It cannot fix total shortage.





Own simulation (500 farmers x 3 seeds): served-in-window baseline 19.7% vs KhadSetu 37.8%; wasted trips 1.31 vs 0.02. Where KhadSetu does not help is listed in the findings. Intake parser 30/30; forecast MAE 3.92 vs naive 4.31; 100 threads on 50 bags: 0 over-reserved.

Limits, said out loud: no fertiliser is created; baseline is our model, not measured data; OTP/Farmer ID/SMS/stock feeds are mocked; forecaster and anomaly watch train on synthetic data until real data exists.